Questions

Q1.

Some questions must be answered with a cross in a box (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

When water boils and turns into steam, there are changes in the arrangement of particles and the density.

Which of these shows the changes?

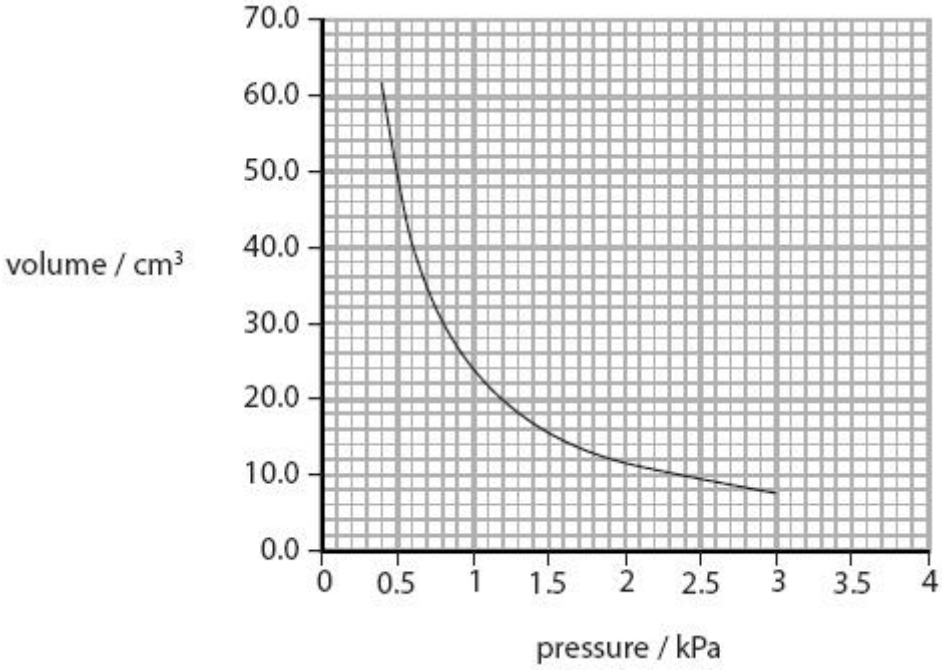
(1)

		space between particles in steam	density of steam
☐	**A**	bigger than in water	greater than water
☐	**B**	bigger than in water	less than water
☐	**C**	smaller than in water	greater than water
☐	**D**	smaller than in water	less than water

(Total for question = 1 mark)

Q2.

A student investigated how the volume and pressure of a gas were related.
The graph shows how the volume of a gas changes with pressure.



The table shows the results used to plot the graph.

pressure / kPa	volume / cm^3
2.5	.....
2.0	11.9
.....	14.0
1.4	17.0
1.0	24.0
0.4	61.5

(i) Use the graph to complete the table.

(2)

(ii) The results were taken at a constant temperature of 23 °C.
Complete the sentence by putting a cross (☒) in the box next to your answer.
A temperature of 23 °C can be written in kelvin as

(1)

☐ **A** −273 K

☐ **B** 250 K

☐ **C** 273 K

☐ **D** 296 K

(iii) Estimate a value for the volume when the pressure becomes 4 kPa.

(1)

volume =.....cm^3

(iv) When the pressure of the gas is 2.2 kPa, the volume of the gas is 10.8 cm^3.

Use the equation

$$V_2 = \frac{P_1V_1}{P_2}$$

to calculate the volume of the gas when the pressure of the gas is 0.2 kPa.

(2)

volume =.....cm^3

Q3.

Room temperature is 20 °C.

What is 20 °C on the kelvin temperature scale?

(1)

- ☐ **A** 293 K
- ☐ **B** 273 K
- ☐ **C** 253 K
- ☐ **D** 120 K

(Total for question = 1 mark)

Q4.

Figure 21 shows a shed made mostly of concrete blocks.

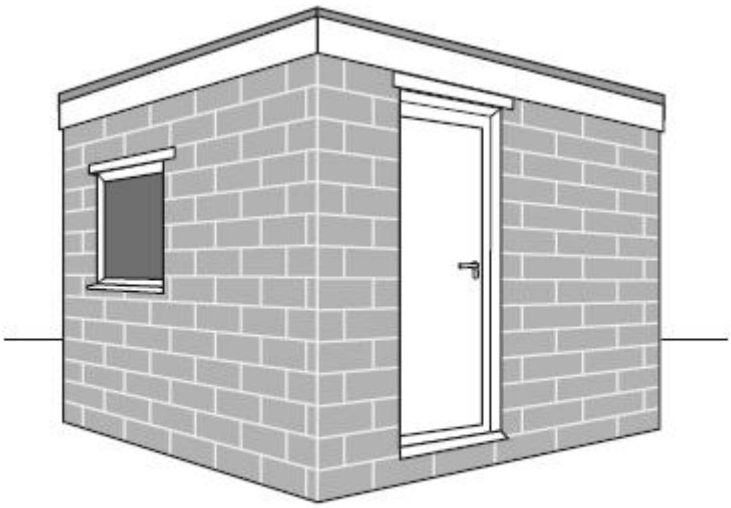


Figure 21

State **two** practical ways to reduce heat loss from this shed.

(2)

1 .....

.....

2 .....

.....

(Total for question = 2 marks)

Q5.

The container of helium is moved from a cold room to a warmer room.

State the effect of an increase in temperature on the helium gas particles inside the container.

(1)

.....

.....

(Total for question = 1 mark)

Mark Scheme

Q1.

Question number	Answer	Mark
	[x] **B** bigger than in water less than water A is incorrect because the density of steam is less than water. C is incorrect because the space between the particles increases. D is incorrect because the space between the particles increases and density of steam is less than water.	(1) AO1

Q2.

	Answer	Acceptable answers	Mark
(i)	volume in range 9.0 – 10.5 (cm^3) (1) pressure in range 1.5 – 1.7 (kPa) (1)		**(2)**
(ii)	☒ **D** 296 K		**(1)**
(iii)	Volume in range 4 – 8 (cm^3)	Any value between 4 (cm^3) and 8 (cm^3)	**(1)**
(iv)	Substitution (1) 2.2 × 10.8 ÷ 0.2 Evaluation (1) 119 (cm^3)	118.8 (cm^3) give full marks for the correct answer, no working	**(2)**

Q3.

Question number	Answer	Mark
	A 293 K	(1)

Q4.

Question number	Answer	Additional guidance	Mark
	statements to include any **two** from use cladding / (extra) insulation (1) use double thicknesses of the concrete (1) use silver / reflective / white (paint) (1) plant trees around (wind break) (1) use double glazed windows (1) (properly) close window(s)/door	 create cavity draft exclusion	**(2)** **AO1**

Q5.

Question number	Answer	Additional guidance	Mark
	particles have greater (kinetic) energy (1)	greater (average) velocity move faster/ more quickly more (frequent) collisions exert greater force on sides ignore changes pressure of the gas	**(1)** **AO1**